

1) Implement a 16:1 MUX using pseudo Nmos logic?

A 16:1 MUX select one of 16 inputs based on 4 select line.

Input: $D_0, D_1, \dots, D_{15}$.

select line: S_3, S_2, S_1, S_0 .

output Y

Boolean expression

$$Y = D_0 \overline{S_3} \overline{S_2} \overline{S_1} \overline{S_0} + D_1 \overline{S_3} \overline{S_2} \overline{S_1} S_0 + \dots +$$

• only term is active for a given select combination

